

Curriculum Vitae
Seth M. Pritchard

(210) 571-6680 seth.p@live.com Red Lion, PA

Education

B.S., Physics, December 2022

University of Texas at San Antonio, San Antonio, TX 78249

Employment

Calibration Technician, October 2023 - September 2025

Intertek, York, PA

Repaired and calibrated measurement devices used in materials research and construction testing. Performed tests to determine uncertainties for various assets. Used statistical methods to ensure validity of uncertainties and measurements during calibrations. Updated existing and wrote new calibration procedures. Performed lab administration duties such as asset shipping/receiving, processing of certs and datasheets, and scheduling testing services.

Undergraduate Research Assistant, August 2017 – May 2019

Department of Physics and Astronomy, UTSA

Studied the oxidation effects of high-temperature steam on silicon carbide in order to better understand structural failure modes in SiC ceramics. Prepared samples for analysis in a simultaneous thermal analyzer. Imaged samples under electron microscopy. Presented at undergraduate conference.

Undergraduate Research Assistant, August 2013 - May 2014

Department of Astronomy and Astrophysics, Penn State University

Assisted in research evaluating the architecture of the planetary system around GJ 876 and searching for evidence of resonances and co-planarity. Wrote python scripts to convert radial velocity data from the Hobby-Eberly Telescope into a format that could be easily parsed by a Markov chain Monte Carlo algorithm. Aided in the preparation of paper for publication.

Publications

Pritchard, S.M. (June 2025) *Lightcurve and Rotation of 55 Pandora*. Paper published in Minor Planet Bulletin.

Nelson, B.E., Robertson, P.M., Payne, M.J., **Pritchard, S.M.**, Deck, K.M., Ford, E.B., Wright, J.T., Howard, T.I. (November 2015) *An empirically derived three-dimensional Laplace resonance in the Gliese 876 planetary system*. Paper published in Monthly Notices of The Royal Astronomical Society.

Presentations

Pritchard, S., Wood, E.S. (October 2018) *Oxidation of Tristructural Isotropic Fuel Forms under Steam-Bearing Accident Conditions*. Poster presented at UTSA COS Research Conference 2018.

Gaspar, M., **Pritchard, S.**, Fernandez, A., Wood, E.S. (October 2018) *Machine Learning Model to Predict Failure Modes of TRISO Particles under High-Temperature Gas-Cooled Reactor Accident Conditions*. Conference presentation at Bulletin of the American Physical Society.

Pritchard, S. (March 2013) *Curiosity on Mars: 200 Sols and Counting*. Presentation at 61st Annual Conference of the American Association of Physics Teachers, Central Pennsylvania Section.

Relevant Coursework

Fundamentals of Astronomy

Introduction to Astrophysics

Cosmology

Electricity & Magnetism

Electrodynamics

Mathematical Physics 1 & 2

Quantum Mechanics 1 & 2

Data Analysis and Visualization (MATLAB)

Honors and Awards

Honor Roll: *Fall 2022*

Phi Theta Kappa: *2012*

Student Leadership Roles

Vice President *Spring 2018 – Fall 2019*

Society of Physics Students, UTSA Chapter